

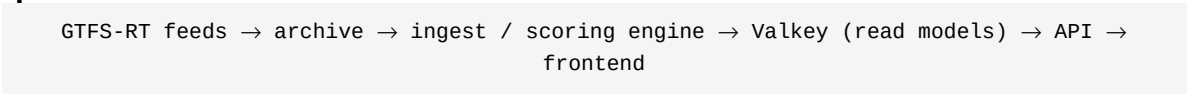
Transit Sentinel turns public transit feeds (GTFS + GTFS-RT) into a **ranked, explainable decision queue**: what's unstable, why it was ranked, what evidence supports it, and what an operator should do next.

It is **not** a CAD/AVL replacement, a rider app, or a generic dashboard. It is an operations intelligence layer between raw telemetry and human decision-making.

*The analogy that lands: Agency alerts are the bulletin board. Transit Sentinel is the triage desk.*

Because we rely entirely on public telemetry, we do not need a formal agency integration to demonstrate value. To prove this, we continuously process **Boston's MBTA public feeds** as our live proving ground at [sepdynamics.co](https://sepdynamics.co) — tracking 192+ routes every 20 seconds, months of continuous operation.

## The Pipeline



## What Makes It Different

| Category               | What's Available Today                                                   | Transit Sentinel Difference                                                                                    |
|------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| CAD/AVL                | Vehicle tracking, schedule adherence, dispatch comms, yard management    | <b>Cross-corridor ranking, explainable scoring, evidence preservation</b> — the triage desk on top of dispatch |
| Raw GTFS-RT feeds      | Raw protobuf/JSON data, requires custom parsing                          | <b>Normalizes, scores, caches</b> — separates public vs. protected API surfaces, adds API products             |
| Performance dashboards | Historical reliability metrics (on-time %, speed)                        | <b>Live current-state triage</b> — answers “what needs attention right now”                                    |
| Rider apps             | Trip planning, ETAs, rider navigation (Transit, Google Maps, Apple Maps) | <b>Built for operators</b> and public-information teams, not riders                                            |

Your CAD/AVL shows: Vehicle #1432    Lat: 34.0522    Lon: -118.2437    +3min delay

▼

Transit Sentinel shows: #1 Red Line    Hazard: 0.81    Evidence: 14min delay, 3 high-impact alerts, headway collapsed → Dispatch relief

## Key Differentiators

- **Public-feed-first deployment:** Works from GTFS-RT feeds agencies already publish. No procurement, no internal data-sharing agreement needed to demonstrate value.
- **Scoring with provenance:** Every ranking carries an evidence trail — not an opaque score. You see *why*.
- **Public/API separation:** `/api/status/*` (rider-facing) and `/api/transit/*` (operations) share the same engine with different auth, caching, and schema.
- **Replayable proof (case packs):** Scoring changes validated against labeled historical incidents. Prove improvement or detect regression before deploying.
- **Performance-engineered architecture:** Materialized read models, Redis pipelines, ETag caching, bounded resources. Processes network-wide telemetry with minimal infrastructure footprint.

## Where the Value Is

| Operations controller                                                                           | Public information team                                                             |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Ranked triage queue — the 8 routes needing attention now, with evidence and recommended actions | Public status page + JSON API with rider-friendly language, feed quality monitoring |

**Data / analytics team**

Replayable proof, case pack calibration, trend analysis, automated scorecard

**IT / integration team**

Clean `/api/status/*` contract, feed health monitoring

**Executive / board**

Live uptime tracking, incident tracking, performance scorecard across all modes

## What We Have Proven Today (Boston Proving Ground)

Transit Sentinel proves that a useful operations intelligence product can be bootstrapped from public transit standards before requiring internal agency integrations.

- **Live Boston proving ground:** We selected Boston to stress-test our engine due to its rich public data availability and complex multi-modal network. We actively process the MBTA's live network at [sepdynamics.co](https://sepdynamics.co) with months of zero-downtime continuous operation.
- **Scale:** 192+ routes tracked, current feed ingest every 20 seconds
- **Verifiability:** Rigorous calibration suite — 6 case packs with 16 labeled incidents (both positive and quiet controls) ensuring scoring changes are auditable, not subjective
- **Resilience:** High-efficiency architecture (Valkey read models, bounded memory limits) processing network-wide telemetry without massive cloud overhead
- **Surface area:** 20+ API endpoints across public and protected surfaces, full test suite, benchmark artifacts, live health monitoring

## Pilot Opportunity

Transit Sentinel's technical foundation is designed to be reusable across any agency that publishes GTFS and GTFS-RT. Because we rely solely on public telemetry for the first wedge, we do not require a heavy procurement cycle or internal dispatch data-sharing agreements to demonstrate value.

**We are currently looking for our first official pilot partner** to run alongside our live Boston tracker. We can focus the pilot on whichever wedge is most immediately valuable to your team:

1. **Live status / API monitoring** — a public status page and JSON API for riders and third-party developers
2. **Operations triage and prioritization** — the ranked, explainable decision queue for your control center
3. **Feed-quality assurance** — continuous monitoring and reporting of your GTFS-RT feed health
4. **Reliability proof reports** — automated scorecards and replayable evidence for service events